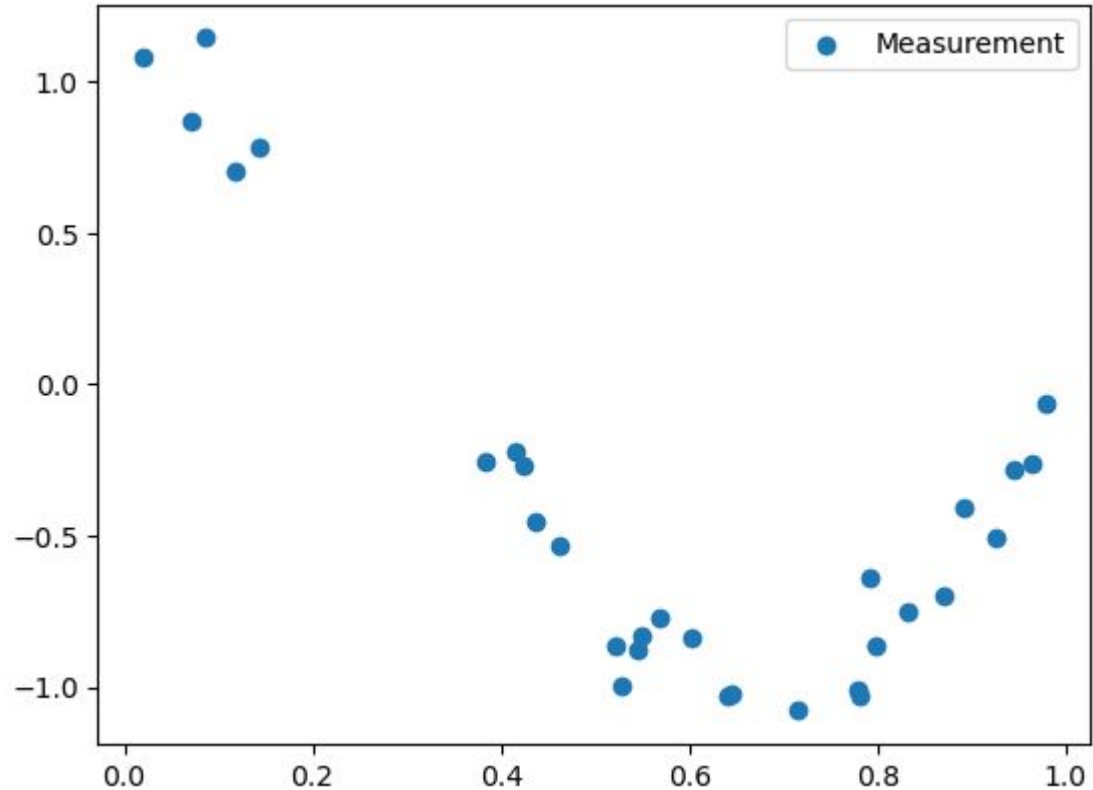


Performance Evaluation Strategies

Overfitting

- Some Measurements
- X is our feature
- Y is what we want to predict
- Train a model that predicts Y



Overfitting

- Linear model

```
1 linear_regression = LinearRegression()
2 linear_regression.fit(X[:, np.newaxis], y)
3
4 y_pred = linear_regression.predict(X[:, np.newaxis])
```

- Some regression performance metrics

```
13 print(f"MAE = {mean_absolute_error(y, y_pred):.2f}")
14 print(f"MSE = {mean_squared_error(y, y_pred):.2f}")
15 print(f"R2 = {r2_score(y, y_pred):.2f}")
```

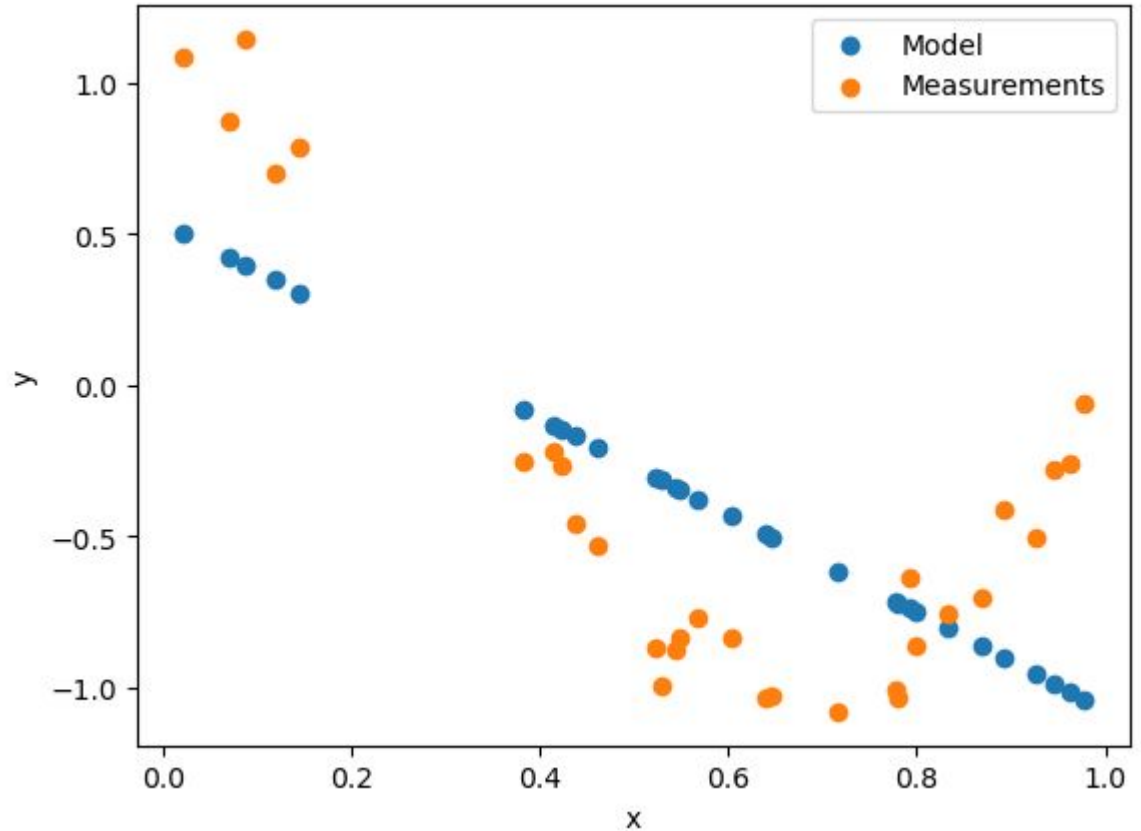
Overfitting

- Linear model
- ...sucks

MAE = 0.42

MSE = 0.23

R2 = 0.47



Overfitting

- More complex model!

```
2  polynomial_features = PolynomialFeatures(degree=degree, include_bias=False)
3  linear_regression = LinearRegression()
4  pipeline = Pipeline(
5      [
6          ("polynomial_features", polynomial_features),
7          ("linear_regression", linear_regression),
8      ]
9  )
10 pipeline.fit(X[:, np.newaxis], y)
11
12 y_pred = pipeline.predict(X[:, np.newaxis])
```

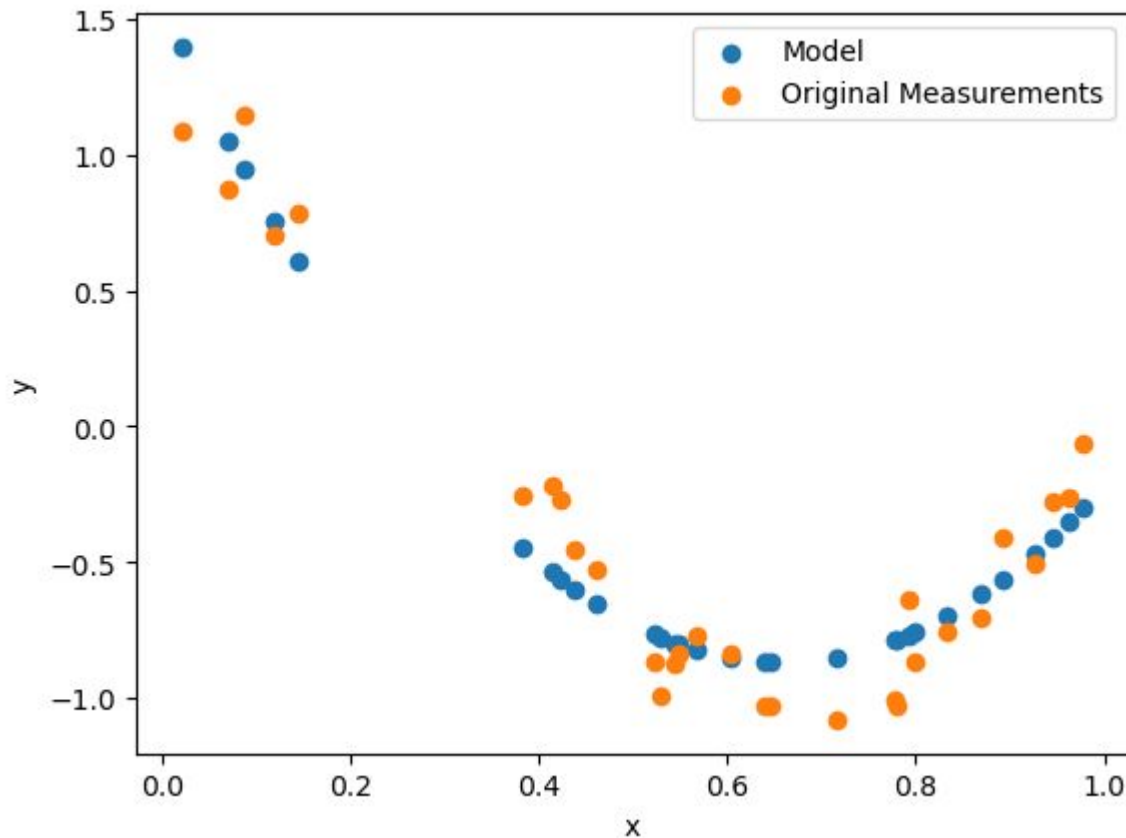
Overfitting

- Polynomial Features
- Degree 2

MAE = 0.15

MSE = 0.03

R2 = 0.93



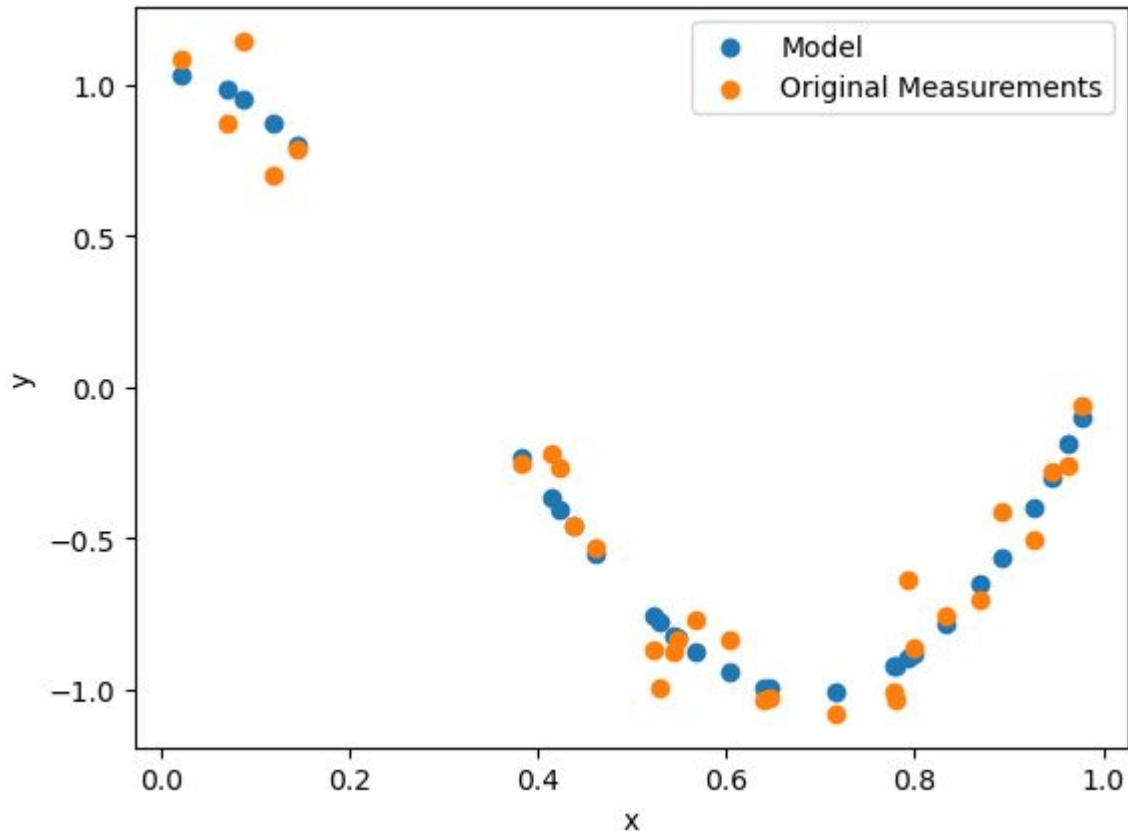
Overfitting

- Polynomial Features
- Degree 4

MAE = 0.08

MSE = 0.01

R2 = 0.97



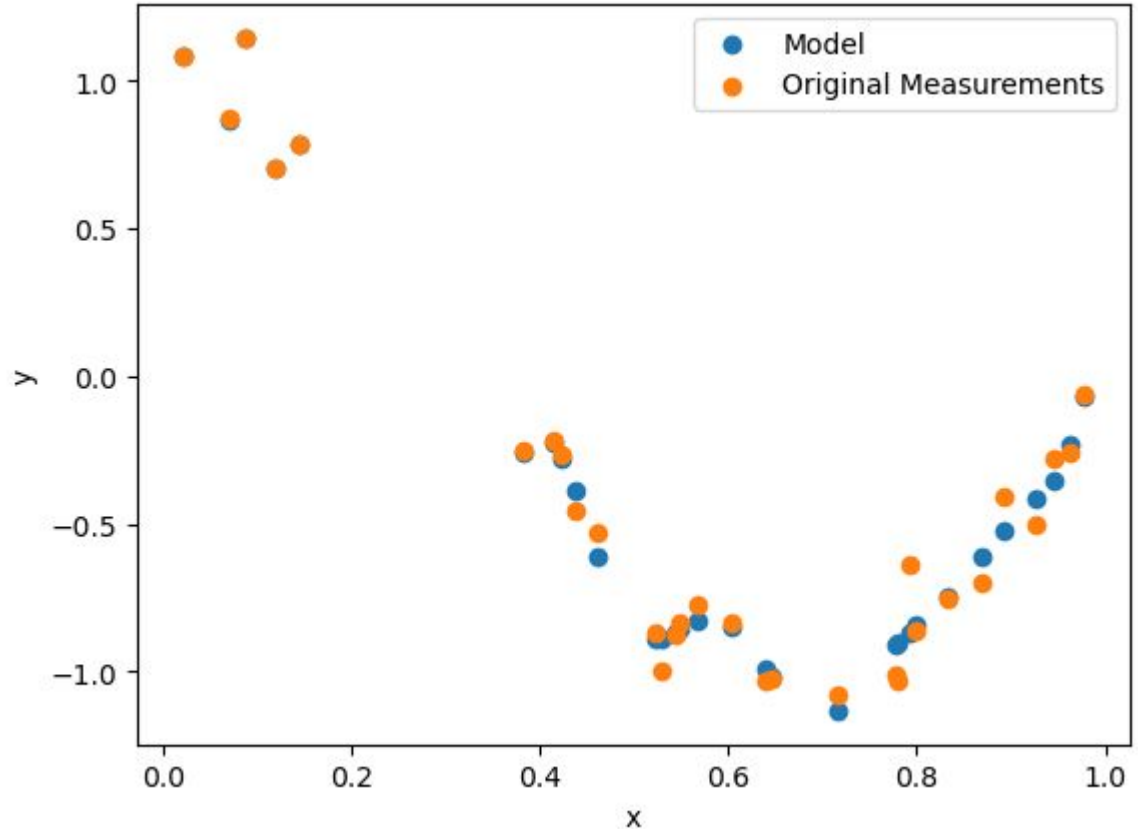
Overfitting

- Polynomial Features
- Degree 16

MAE = 0.05

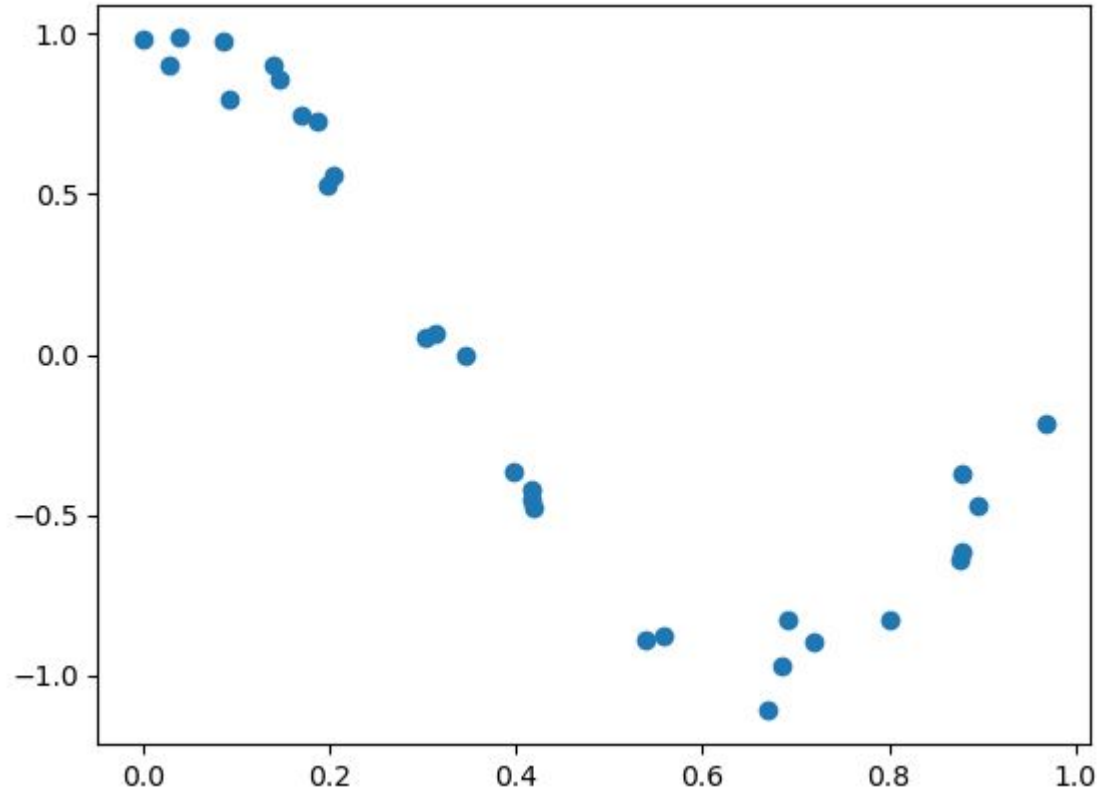
MSE = 0.00

R2 = 0.99



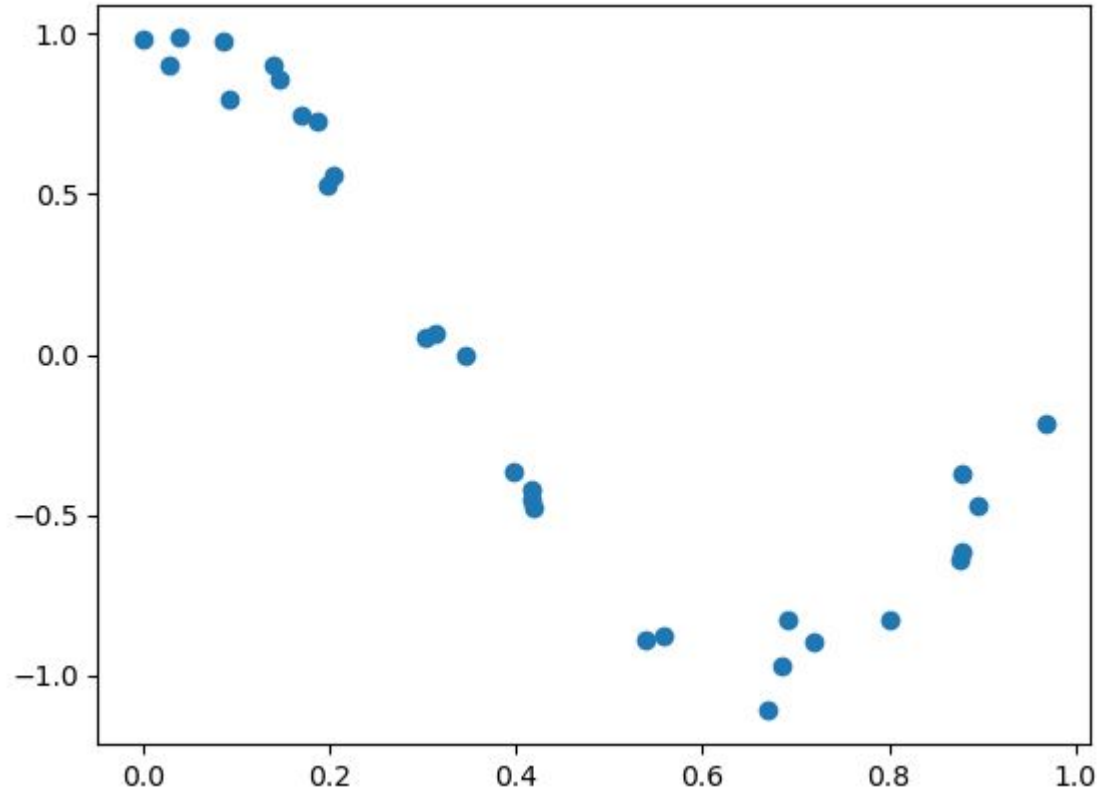
Overfitting

- Reality
- We get new measurements



Overfitting

- Reality
- We get new measurements



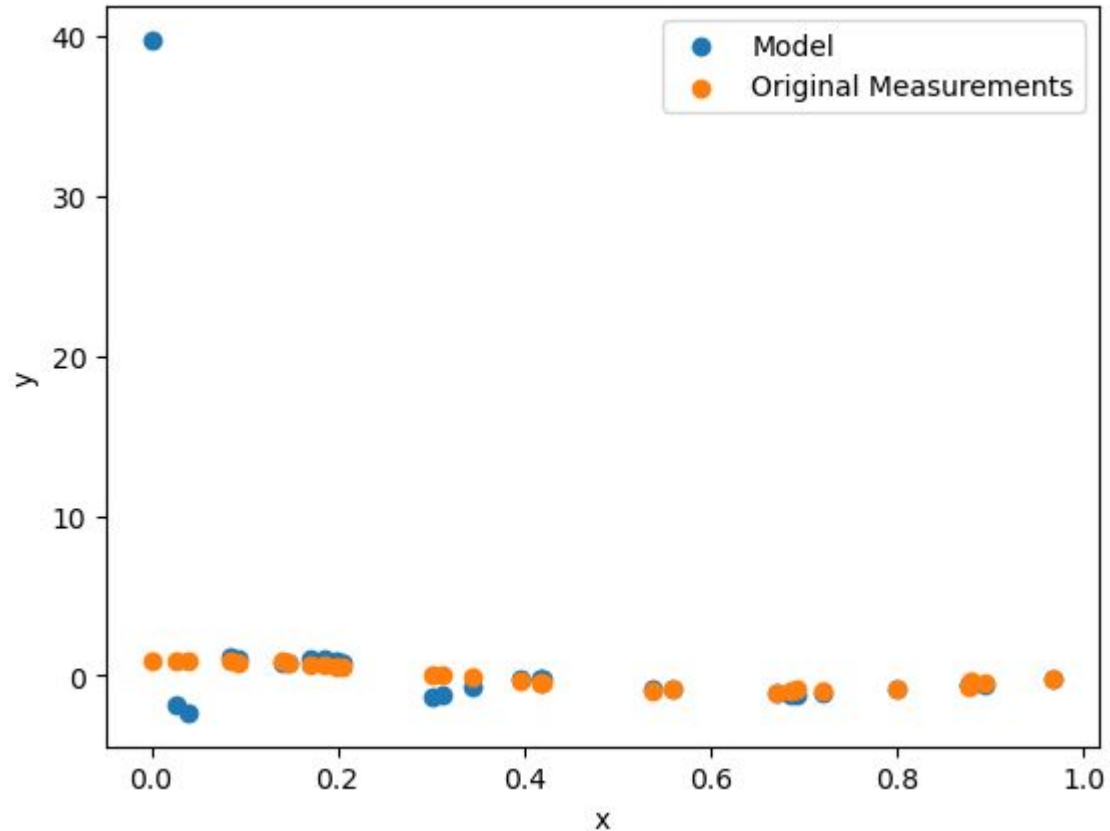
Overfitting

- Our best model from before

MAE = 1.74

MSE = 50.83

R2 = -98.13



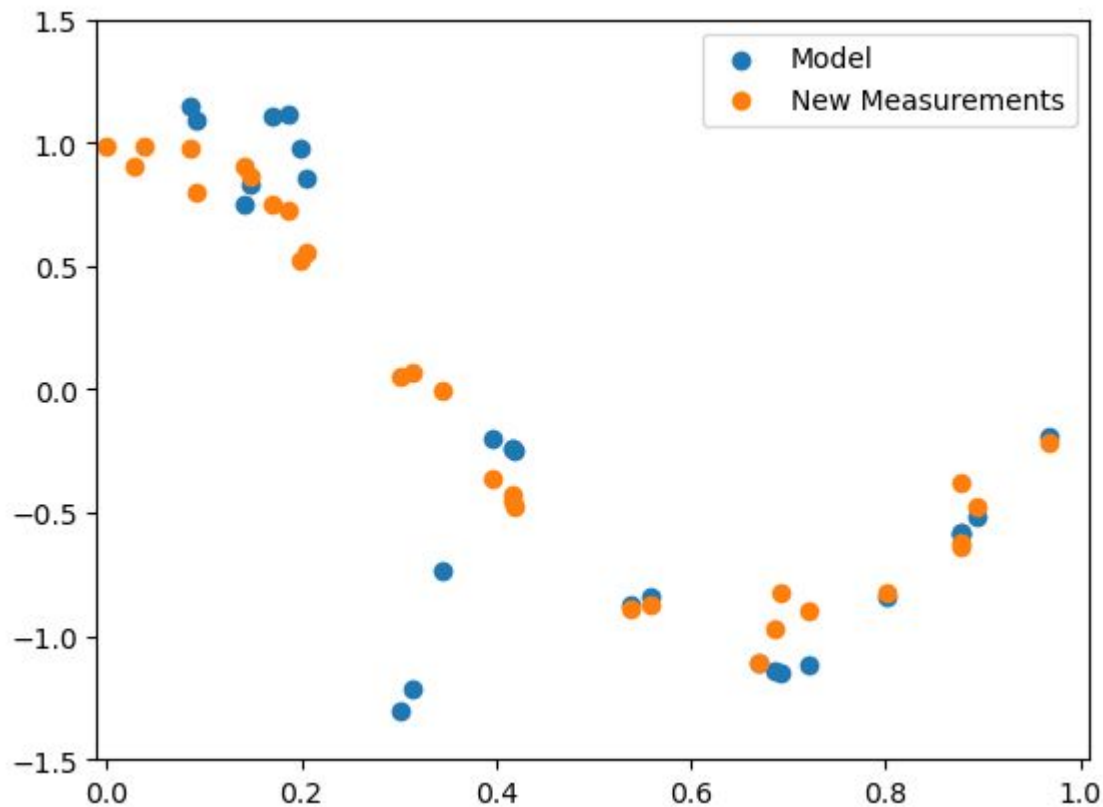
Overfitting

- Our best model from before
- Zoom in a bit

MAE = 1.74

MSE = 50.83

R2 = -98.13



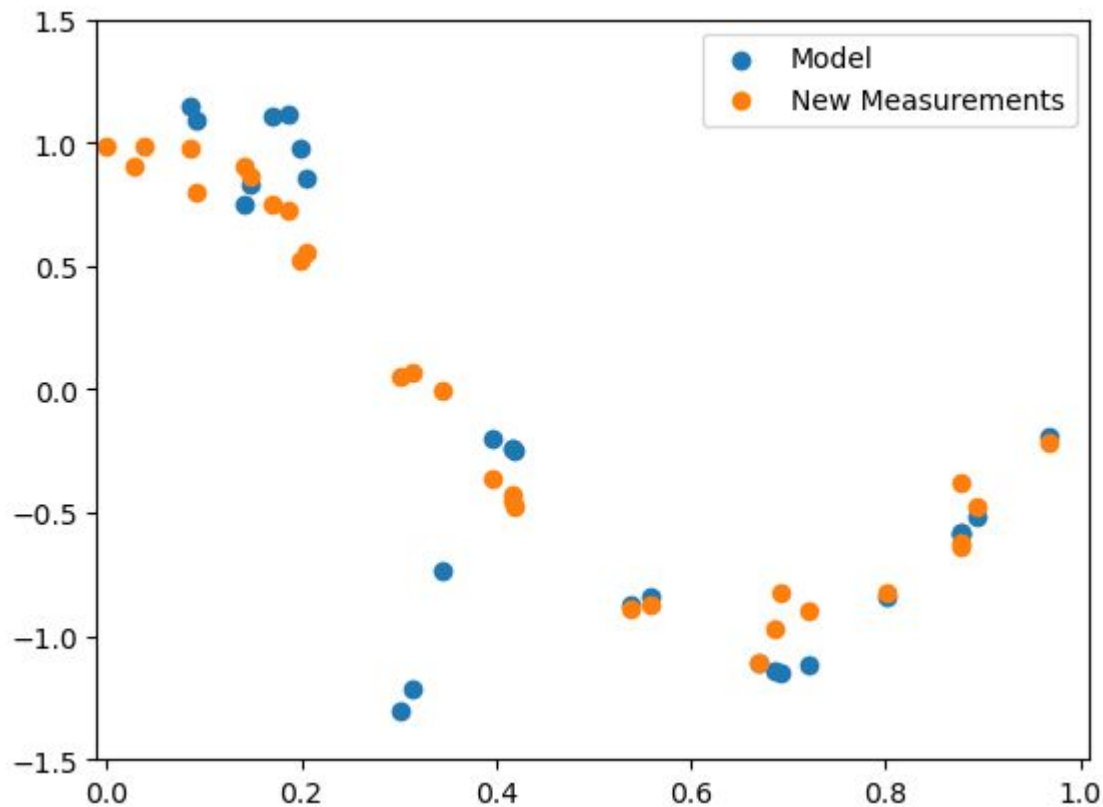
Overfitting

- Our best model from before
- Zoom in a bit

MAE = 1.74

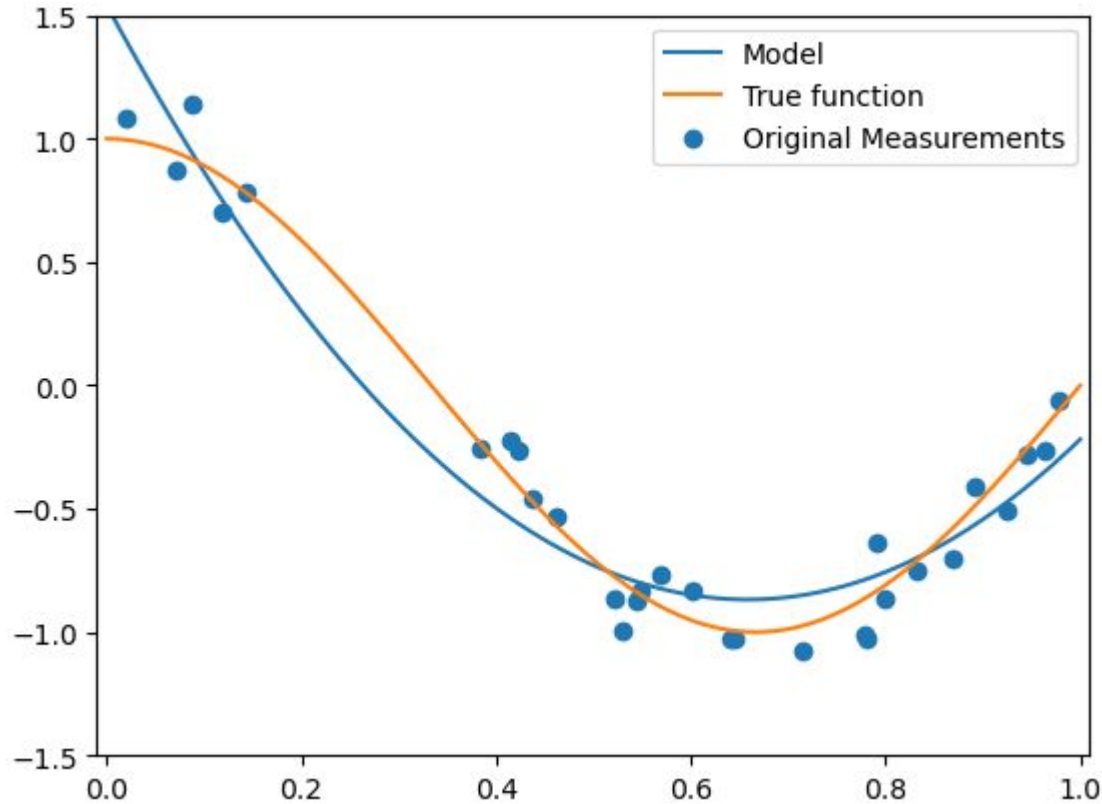
MSE = 50.83

R2 = -98.13



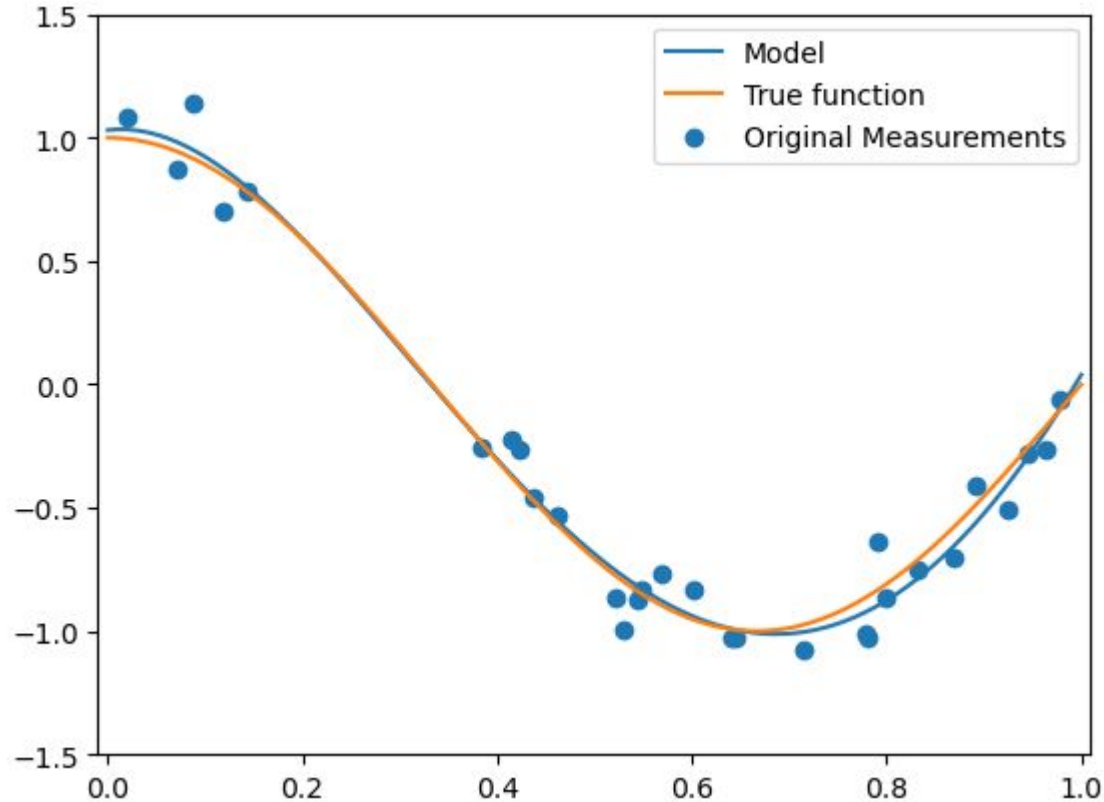
Overfitting

- What we normally can't see
 - The function we learned
 - The function we wanted
-
- Polynomial features
 - Degree 2



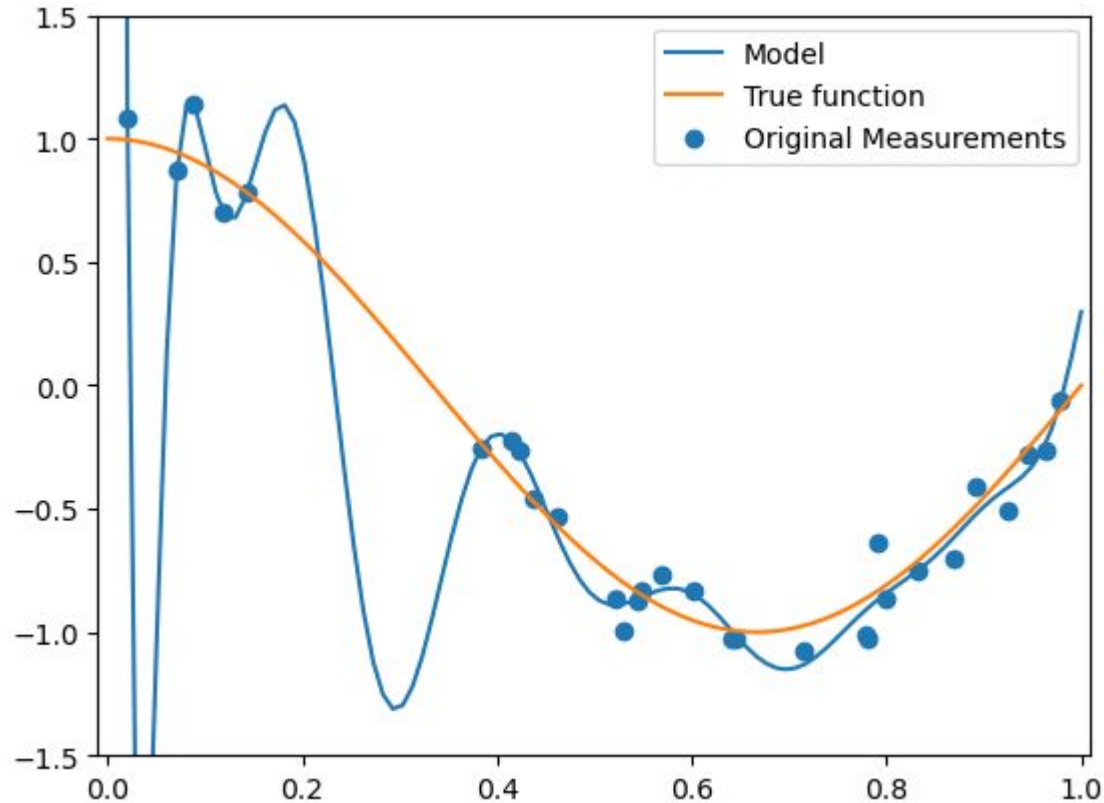
Overfitting

- What we normally can't see
 - The function we learned
 - The function we wanted
-
- Polynomial features
 - Degree 4



Overfitting

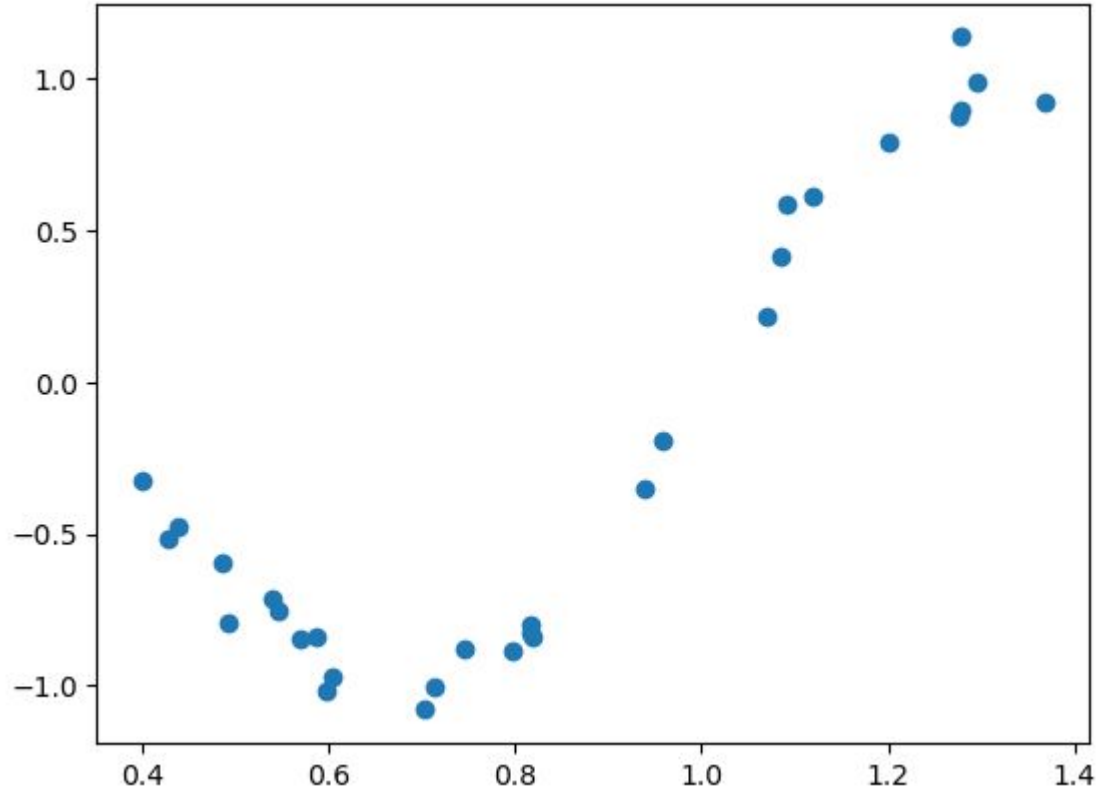
- What we normally can't see
 - The function we learned
 - The function we wanted
-
- Polynomial features
 - Degree 16



More problems

Out of distribution

- Again, reality
- We get new measurements
- They go beyond the values found in the training set



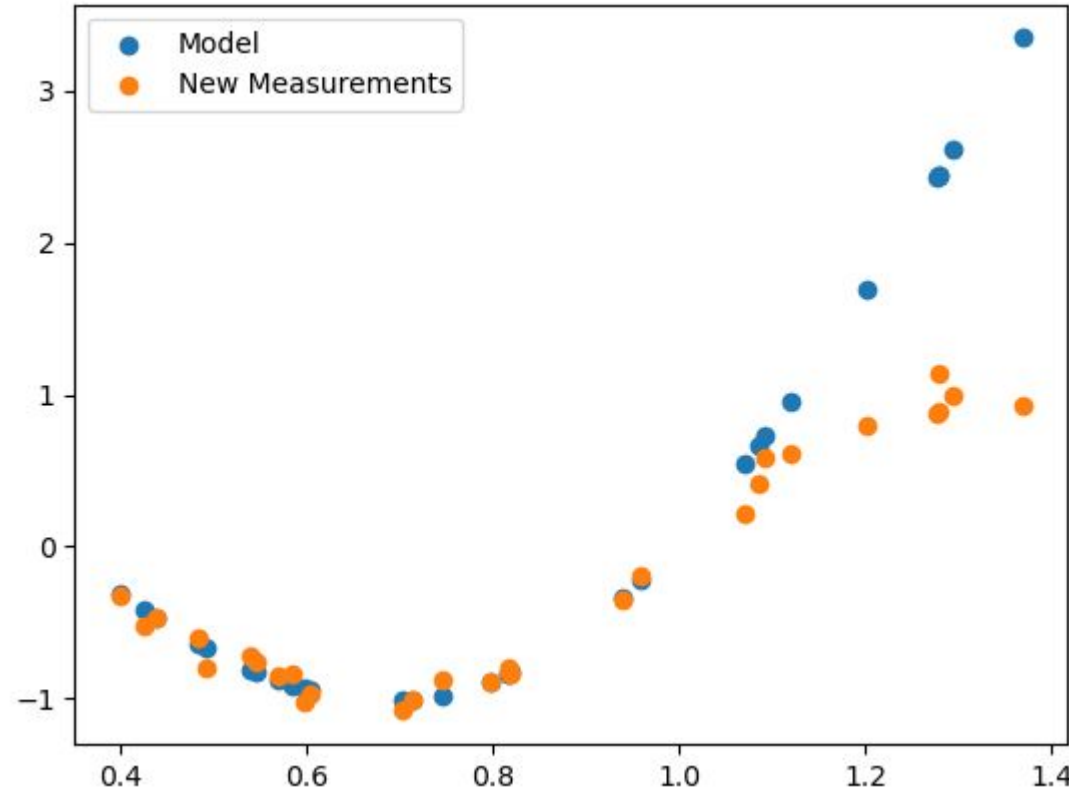
Out of distribution

- Best model we saw in our Vizualisation above
- Polynomial features
- Degree 4

MAE = 0.38

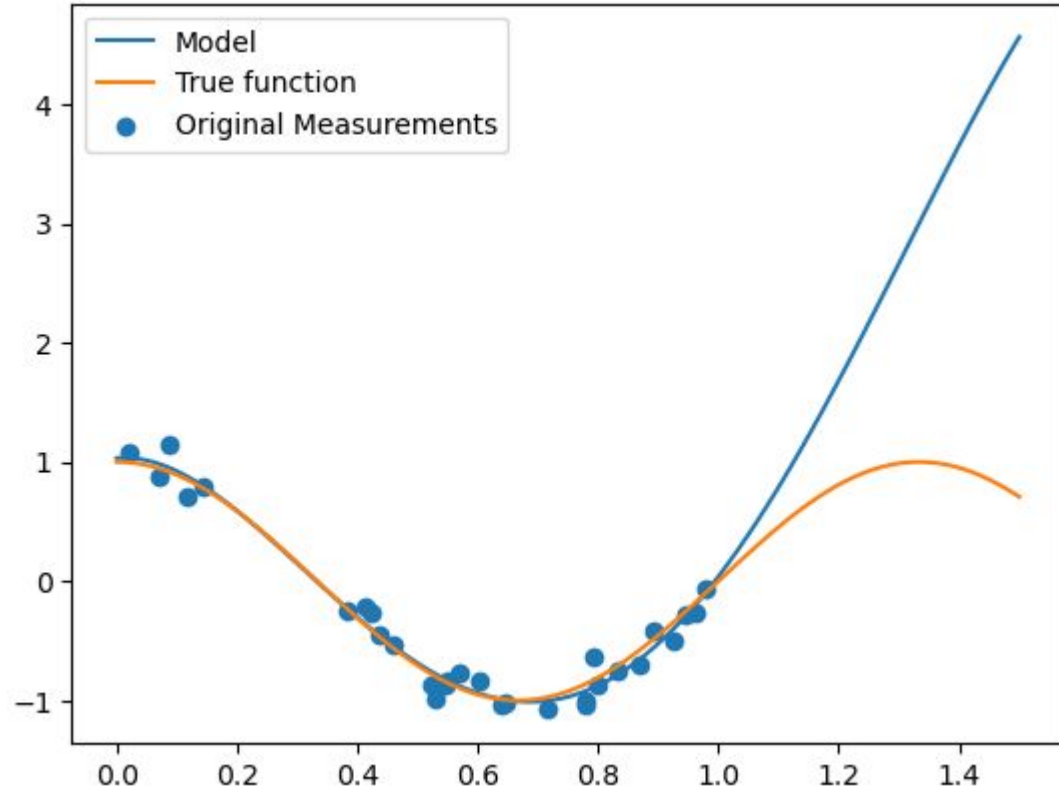
MSE = 0.54

R2 = 0.01



Out of distribution

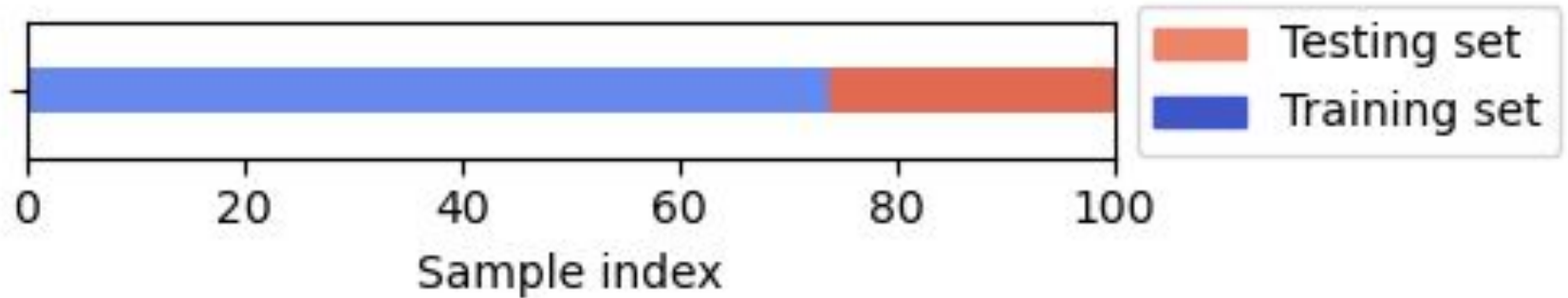
- What we normally can't see
 - The function we learned
 - The function we wanted
-
- Polynomial features
 - Degree 4



Ideas on how to measure model performance

Train-Test Split

- Split the whole dataset
- Train on one part
- Evaluate on the other

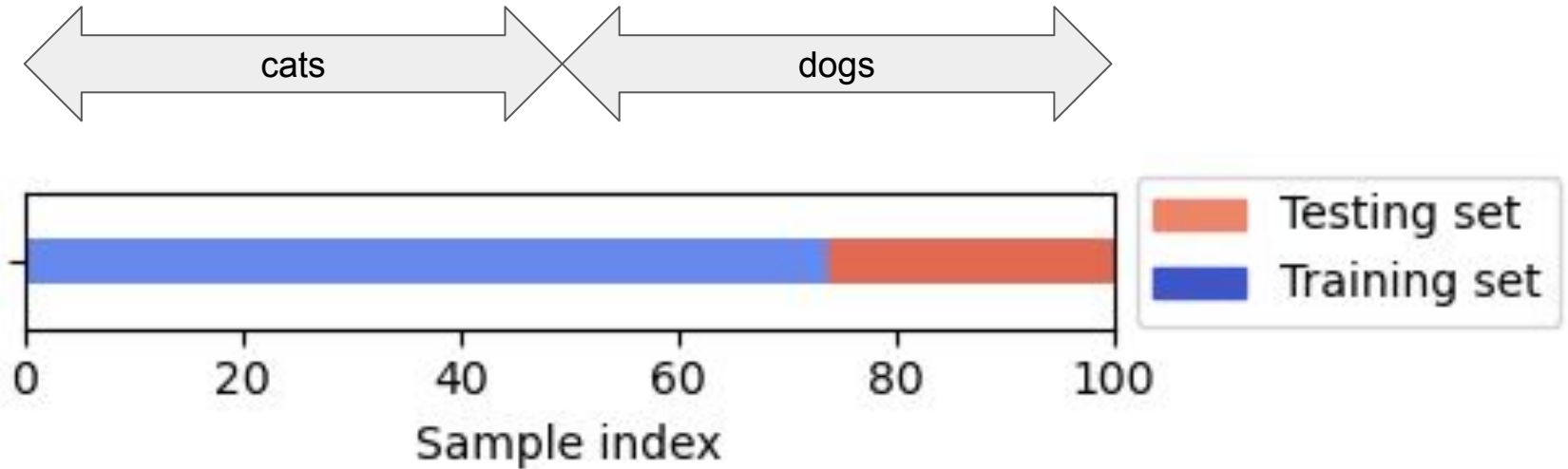


Train-Test Split - Problems

-

Train-Test Split - Problems

- Order



Train-Test Split - Problems

- Order



Train-Test Split - Problems

- Order
- Solution: Shuffle before splitting!

Train-Test Split - Problems

- Order
- Solution: Shuffle before splitting!
- unless ... Timeseries data! (more on that later)

Train-Test Split - Problems

- Produces a single evaluation score
- Maybe test set ended up being “easy” by chance?

Ideas on how to get more confident in results?

Repeated shuffled train test split

- Repeat the shuffling and train test splitting?
- Provides more performance measurements.

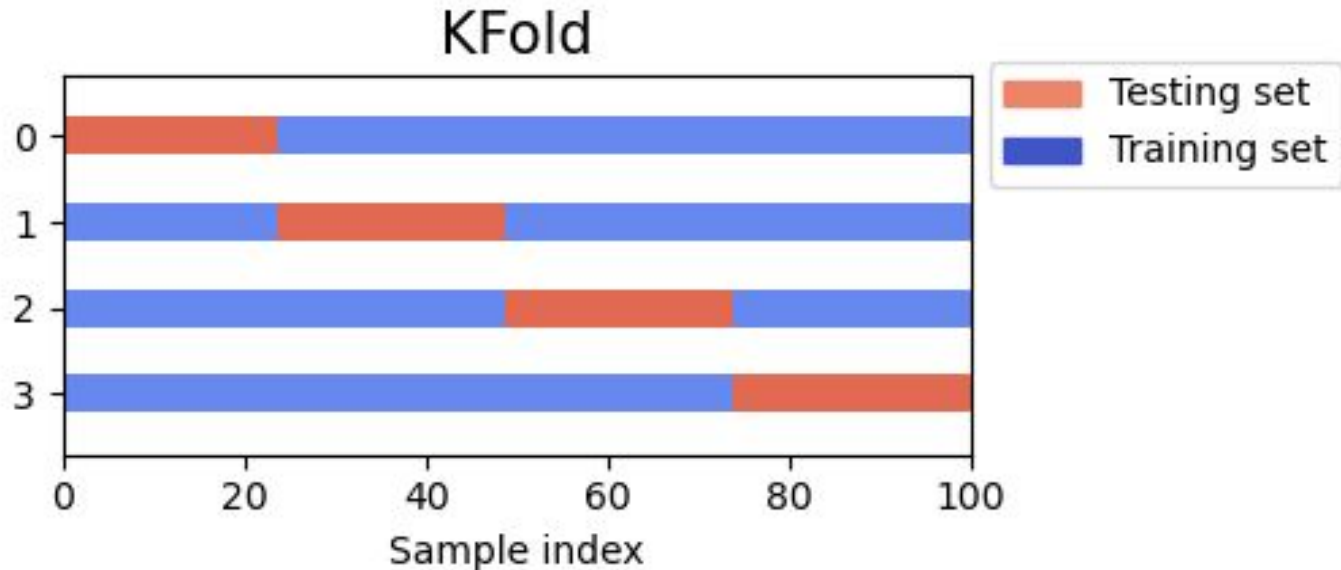
Repeated shuffled train test split - Problems

- Some data may never be in the test set
- Some data may end up in multiple test sets
- Strong overlaps possible
- Problematic for statistical significance testing
- Computationally expensive
- (Not useful for timeseries data)

Better Ideas on how to get more confident in results?

k-Fold Cross Validation

- Split the whole dataset in k chunks
- Train on n-1 parts
- Evaluate on the one leftover
- Repeat n times



k-Fold Cross Validation

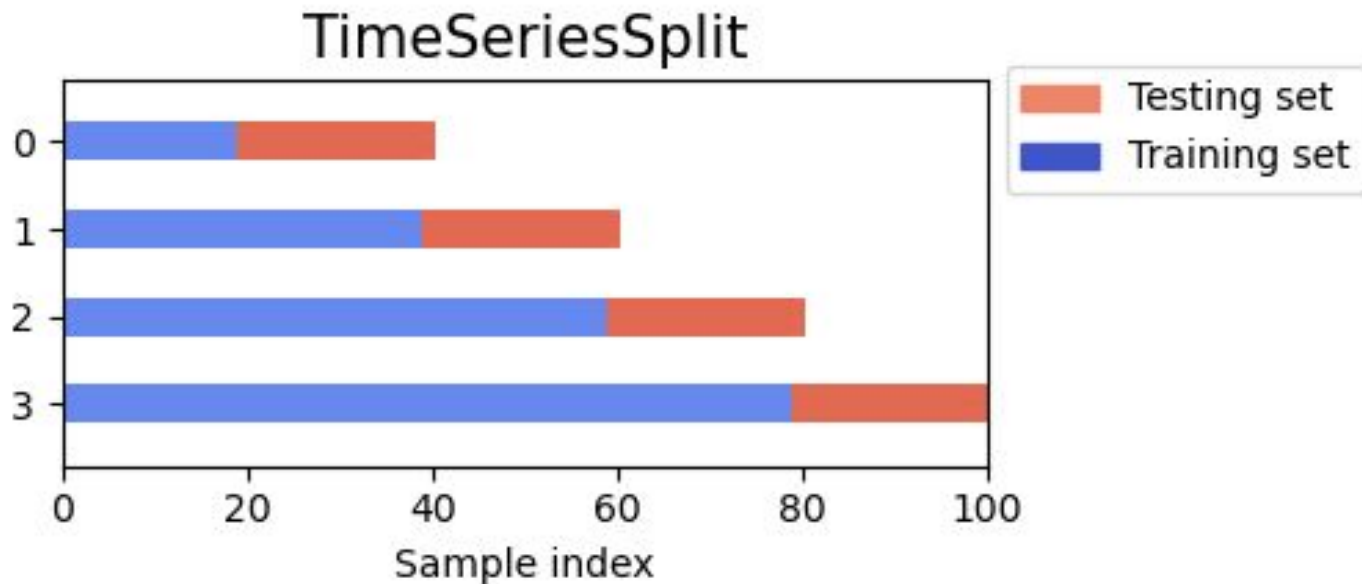
- Produces k scores
- Widely applied in practice
- Allows for limited statistical significance testing
- If performance scores vary a lot
 - Model has problems
 - Something going on in the dataset

k-Fold Cross Validation - Problems

- Only k scores
- Actually allows only for very limited statistical significance testing
- Not useful for time series data
- Same problems with order as before

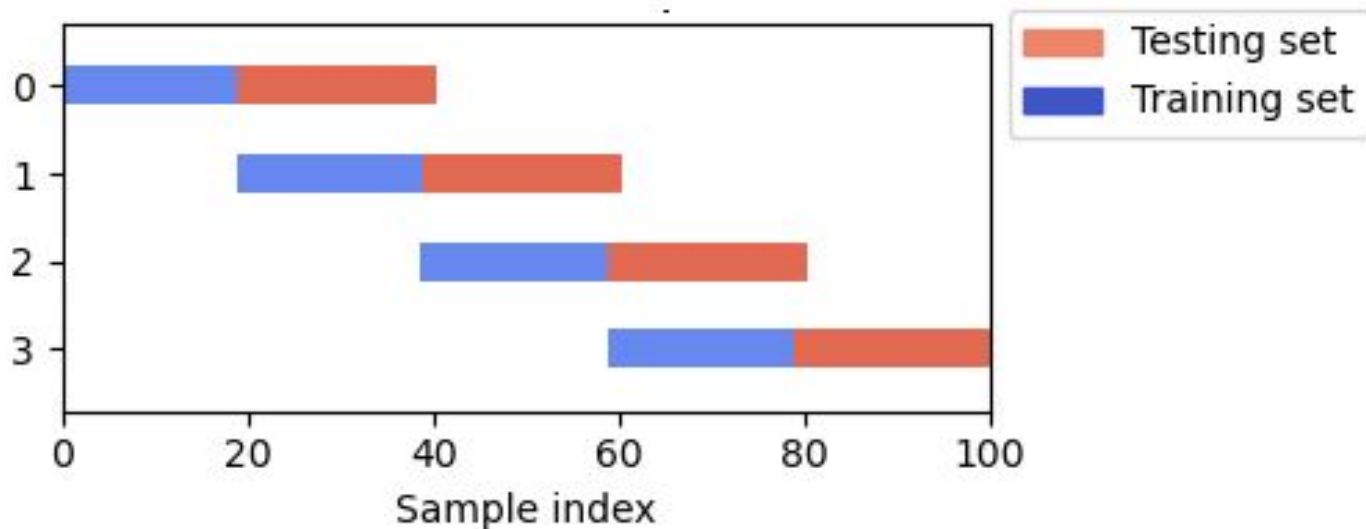
Time Series k-Fold Cross Validation

- Train on first fold, evaluate on the second
- Train on first two folds, evaluate on the third
- ...



Blocked Time Series k-Fold Cross Validation

- Train on first fold, evaluate on the second
- Train on second fold, evaluate on the third
- ...



(Blocked) Time Series k-Fold Cross Validation - Problems

- Requires a lot more data than regular k-Fold CV
- Only k scores
- Allows only for very limited statistical significance testing

Ideas how to produce more performance measurements?

Bootstrap (applied to ML context)

- (Many variations, this is just one of them)
- Repeated train-test split
- Using resampling with replacement
- N times
- Resampling with replacement
 - Items are randomly selected
 - But put back in the sample space
 - The same item can be picked multiple times

Bootstrap (applied to ML context)

- Produces N scores
- Works well for small datasets
- Allows some statistical significance testing

Bootstrap (applied to ML context) - Problems

- Computationally expensive
- Allows only for limited statistical significance testing

Notes on Statistical Significance Testing

- We have a null hypothesis
 - E.g. H_0 = Model A performs better than Model B
- We want a p-value
 - Probability that we see our H_0 true on our data by chance
- Significance $\neq$ Effect size

Parametric Tests

- Student's t-test, z-test, wald-test

Parametric Tests - Student's t-test - Assumptions

- Student's t-test
- Values come from normal distribution (do they?)
- Values have similar variance
- Values are independent (are they?)

Parametric Tests - Student's t-test (don'ts)

- Student's t-test
- Model comparison example
- K scores from each model (e.g. from k-fold CV)
- Treat each as random collection

Parametric Tests - Student's t-test (do)

- **Pairwise** Student's t-test
- Model comparison example
- K scores from each model (e.g. from k-fold CV)

- Test for normal distribution first
- Tests the difference of those K scores
- Be honest on the downsides and issues using student's t-test
- Evaluate effect size

Non-Parametric Tests

- Sampling free
 - Sign test
 - Wilcoxon signed rank
 - Mann-Whitney u-test
 - ...
 - Little statistical “power” but easy and cheap to apply
- Sampling based
 - (paired) Bootstrap
 - Randomization test
 - ...
 - Stronger, but computationally expensive

Take away message

- Use at least k-fold CV unless you have a huge test set
- Be aware of information leakage
- Use time series splits whenever your data contains causal effects
- Random repetition does not increase the statistical power
- Use student's t-test with caution
- Try Bootstrap if theres little data (small data set)
- There is no one golden bullet for statistical signifiance testing